

Lentigo Simplex and Lentiginosis

Lentigines are light to dark brown macules with diameters ranging from $< 1\text{ mm}$ to several centimeters. They commonly appear in childhood but can develop at any age. Mucosal and nail lentigines are more prevalent in individuals with darker skin types. Lentigines may occur sporadically or be inherited as part of genetic syndromes such as LEOPARD syndrome, Peutz-Jeghers syndrome, or Carney complex. They frequently arise on sun-exposed areas, leading to cosmetic concerns in many patients. However, it is essential to exclude a syndromic association before initiating treatment.

Treatment options include topical skin-lightening agents, chemical peels, cryotherapy, and laser or light-based therapies.

Laser and Light Treatment

Numerous randomized controlled trials with blinded assessments have compared laser-treated areas to controls, conventional treatments, or other light modalities for lentigines. These studies generally report favorable outcomes and low recurrence rates—particularly when compared to the treatment of café au lait macules.

Q-switched (QS) lasers—including Q-switched neodymium:yttrium-aluminum-garnet (QSNY), Q-switched alexandrite (QSAL), and Q-switched ruby lasers—are highly effective due to their nanosecond pulses that selectively target melanin. Other modalities such as long-pulsed (LP) alexandrite, pulsed dye laser (PDL), LP Nd:YAG, ablative lasers, and intense pulsed light sources (IPLS) have also been used with variable success.

A newer 600 nm YAG laser has been introduced specifically to reduce the risk of hyperpigmentation and superficial vascular damage. However, a randomized controlled trial did not demonstrate a significant advantage over existing options. Quasi-continuous lasers may serve as second-line therapy for pigmented lesions but should be used cautiously in darker skin types due to increased risk of complications.

Pre-treatment cryotherapy may enhance results but is not recommended as first-line therapy—especially in individuals with darker skin—due to the risk of post-inflammatory hyperpigmentation. The 595 nm PDL can effectively treat lentigines when used with a compression handpiece to minimize purpura. IPLS are also effective for superficial pigmented lesions, producing gradual, uniform epidermal injury while sparing deeper structures. Low-intensity pulses are advised to prevent non-specific epidermal damage. When properly administered, IPLS can achieve a 50% or greater reduction in pigmentation without adverse effects.

One to two treatment sessions are typically sufficient for complete lesion clearance.

Refer to Table S1 (Supporting Information) for a summary of available data on laser treatment of lentigines.

Recommendation

Lentigo simplex is an excellent indication for Q-switched lasers. In cases of multiple lentigines, a syndromic association must be ruled out prior to treatment.

Ephelides (Freckles)

Ephelides, commonly known as freckles, are small, reddish to light brown macules with irregular but well-defined borders. They are typically uniform in size and distribution, become more prominent after sun exposure, and often fade during winter months. Freckles commonly appear on the face, neck, chest, and arms, usually in childhood, and may partially regress with age. They are most frequently observed in individuals with fair skin phototypes, particularly those with red or blond hair.

Sun protection is crucial, as regular use of broad-spectrum sunscreen can significantly inhibit freckle development and help maintain treatment results. Therapeutic options include topical bleaching creams, chemical peels, and laser removal.

Laser and Light Treatment

Freckles are associated with variants in the MC1R gene, resulting in reduced function and a shift from eumelanin to pheomelanin production. Since pheomelanin has an absorption spectrum shifted toward shorter wavelengths compared to eumelanin, the 532 nm wavelength is optimal for targeting freckles.

Short-pulsed pigmented lasers, particularly Q-switched lasers, are considered the gold standard for freckle treatment. These lasers are highly effective and associated with minimal complications, especially in light-skinned individuals.